UNIT 3 · TOPIC 2(B)

Testing of Tillage Equipment

Syllabus topic: Methods of testing and performance evaluation of tillage equipment, seed drills and planters

Checks and laboratory work

- Verify specifications (dimensions, mass, materials).
- Check the hardness of soil-engaging parts (share/point hardness, e.g., ~**350–450 BHN** zones), welds and finish.
- Use soil-bin studies for tool geometry (draft vs rake angle, speed) under controlled soil conditions.

Field performance test

Record the operating conditions **first**:

- Soil type; moisture content (oven method); bulk density (core sampler); cone-index profile (**penetrometer**); surface residue.

Then measure machine performance:

Parameter	How it is measured
Draft	Drawbar pull dynamometer / strain-gauge transducer between tractors or on the hitch (Unit 2, Topic 5)
Speed	Marked run or radar
Working depth & width	Steel scale / profilometer
Wheel slip	Slip % = (1 - advance per rev loaded ÷ advance per rev unloaded) × 100
Fuel consumption	L/h and L/ha — volumetric refill or flowmeter
Field capacity & efficiency	See Topic 6a

Quality of work

- **Pulverization:** clod Mean Weight Diameter (MWD) by sieve analysis — Σ(mass fraction × mean sieve size); a smaller MWD means a finer tilth.
- **Soil inversion and trash burial %** (mouldboard).
- **Furrow cross-section** by profilometer; depth uniformity.
- **Weed kill %** for cultivators/harrows — count weeds before and after.

Worked example — MWD. Clods retained: 50 mm sieve 20%, 25 mm 30%, 10 mm 35%, <10 mm 15% (mean 5 mm) → MWD = 0.2×50 + 0.3×25 + 0.35×10 + 0.15×5 = 10 + 7.5 + 3.5 + 0.75 ≈ **21.8 mm**.

Practice MCQs

Q42. Soil strength during tillage testing is characterised by:

(a) pH meter
(b) Cone index from a penetrometer
(c) EC meter
(d) Altimeter

Q43. Furrow cross-section and surface roughness are measured with a:

(a) Planimeter
(b) Profilometer
(c) Pyranometer
(d) Psychrometer

Q44. Degree of soil pulverization is expressed by:

(a) Bulk density only
(b) Clod Mean Weight Diameter from sieve analysis
(c) Soil colour
(d) Porosity

Q45. Clod fractions: 20% on 50 mm, 30% on 25 mm, 35% on 10 mm, 15% at mean 5 mm. MWD ≈

(a) 12 mm
(b) 18 mm
(c) 21.8 mm
(d) 30 mm

Q46. A finer tilth corresponds to:

(a) Larger MWD
(b) Smaller MWD
(c) Constant MWD
(d) No MWD

Q47. Wheel slip is computed from:

(a) Fuel use
(b) Advance per revolution loaded vs unloaded
(c) Engine rpm only
(d) Tyre pressure

Q48. Trash burial (inversion) evaluation applies mainly to testing of:

(a) Sprayer
(b) Mouldboard plough
(c) Thresher
(d) Duster

Answers

Q42 — (b) Cone index from a penetrometer

Q43 — (b) Profilometer

Q44 — (b) Clod Mean Weight Diameter from sieve analysis

Q45 — (c) 21.8 mm — *MWD = Σ(fraction × mean size) = 0.20×50 + 0.30×25 + 0.35×10 + 0.15×5 = 21.75 ≈ 21.8 mm.*

Q46 — (b) Smaller MWD

Q47 — (b) Advance per revolution loaded vs unloaded — *Slip % = (1 - distance per rev under load ÷ distance per rev unloaded) × 100; measured over a marked run counting wheel revolutions.*

Q48 — (b) Mouldboard plough

Sources: RNAM (UN-ESCAP) Test Codes and Procedures for Farm Machinery; BIS Indian Standards for seed drills, sprayers and dusters; Kepner et al. (performance concepts); Jagdishwar Sahay; FMTTI (Budni) test procedures; NPTEL lectures.